

SME lending starts with understanding financial health

BEFORE · FRAGMENTED INFORMATION

Manual review

Bank Data: cash inflows and outflows, revenue consistency, transaction patterns, business liquidity

GST Data: GST-reported sales, GST paid, filing status

Manual Review + Separate Information: unusual transactions, unexpected credits or debits, irregular behaviour

Bank and GST information can arrive separately, leaving the underwriter to assemble one coherent picture.

→ Underwriter interprets everything

AFTER · UNIFIED ASSESSMENT

Consent-first workflow

Consent

↓

Unified Financial Data

↓

Automated Analysis

↓

Explainable Assessment

Core distinction: The platform does not issue final credit approvals; it provides a transparent, explainable financial health assessment that augments the human underwriter.

A consent-first workflow turns authorized data into one assessment

LANE 01 · BANK PORTAL

Lender requests access

Loan officer initiates loan file and parameters

Select business profile: Choose SME profile (e.g., demo scenario SME001)

Enter loan terms: Specify requested facility (₹8,00,000) and stated purpose

Issue consent request: Generates unique consent ID in PENDING status

LANE 02 · SME PORTAL

Borrower reviews and decides

Business owner inspects requested scope and acts

Inspect scope: View requesting institution, requested tenure, and data types

Data perimeter: Confirms access strictly to bank statements and GST filings

Explicit decision: SME submits APPROVE or REJECT; rejected requests halt immediately

CONVERGENCE GATE

System execution

Consent is the strict prerequisite gate: the backend cannot fetch Account Aggregator records or run the deterministic risk engine while consent remains PENDING or REJECTED. Only an APPROVED token unlocks data retrieval.

System design rule: Consent token validation executes before data retrieval triggers. Zero data persists without prior approval.

From loan application to explainable risk assessment

01	02	03	04	05	06
Application STAGE 01 Synthetic demo profile SME001; requested amount ₹8,00,000 and loan purpose.	Consent STAGE 02 Consent request created with status PENDING.	Approval STAGE 03 · GATE SME chooses APPROVE or REJECT; lifecycle: PENDING → APPROVED or PENDING → REJECTED.	Retrieval STAGE 04 After approval, the Mock Account Aggregator returns approximately 12 months of synthetic bank and GST history.	Analysis STAGE 05 Deterministic engine evaluates cash flow, revenue, debt-burden proxy, GST compliance, and transaction risk.	Review STAGE 06 Dashboard shows score, metrics, flags, explanations, and a preliminary recommended credit limit.

DEMO PROFILE CONTEXT · SME001

SME001 is a synthetic demo profile, not a real customer.

STATE MACHINE INVARIANTS

PENDING → APPROVED

PENDING → REJECTED

The architecture keeps Bank and SME experiences behind one backend

FRONTEND LAYER · NEXT.JS

Unified client app

Deployed on Vercel

Bank Portal

Application setup & review

SME Portal

Consent inspection & decision

CORE BACKEND · FASTAPI

API & orchestration

Deployed on Render (Python)

Application & consent lifecycle

REST routes managing state

Deterministic risk engine

Rule evaluation & scoring

In-memory session state

Non-persistent storage

DATA LAYER · MOCK PROVIDER

Account Aggregator

Simulated API provider

Synthetic bank feed

12-month transaction ledgers

Synthetic GST returns

Monthly sales & tax paid

Consent verification

Enforces approved token check

Architectural isolation note: The Bank Portal and SME Portal do not communicate directly. All cross-party operations, consent state transitions, data fetching, and underwriting outputs route strictly through the centralized FastAPI backend layer.

Stack specification: Next.js (TypeScript) on Vercel · FastAPI (Python) on Render · In-memory state store · Simulated AA provider

Weighted signals turn synthetic financial data into an explainable assessment

SCORING FRAMEWORK WEIGHTS

Component	Weight
Cash Flow	35%
Revenue	20%
Debt Health	25%
GST Compliance	10%
Transaction Risk	10%

DATA EVALUATED

Bank data: transaction ID, date, description, amount, credit/debit type, balance, category

GST data: monthly reported sales, GST paid, filing status

RECOMMENDATION FORMULA

Credit limit calculation

$$\text{Recommended Credit Limit} = \text{Requested Amount} \times \text{Overall Score} / 100$$

At ₹8,00,000 and 72/100, the preliminary recommendation is:

₹5,76,000

Preliminary recommendation only — not an approved loan amount.

RULE-BASED EXAMPLES

TXN044: “Unregistered Crypto Exchange Transfer” for ₹3,50,000 DEBIT

TXN048: “Massive Uninvoiced Wire Deposit” for ₹6,00,000 CREDIT

Explainability chain: Metric Scores → Overall Score → Risk Flags → Human-readable Explanation → Preliminary Recommended Credit Limit

Boundary: Debt Health is a synthetic debit-burden proxy, not a complete debt schedule; the engine is deterministic, not a production fraud-detection model.

The user journey ends in a structured review, not a black-box decision

SCHEMATIC 01 · BANK PORTAL

Application setup

Lender initialization interface

Applicant: SME001 Retail Demo

Requested: ₹8,00,000 (Working Cap)

Action: [Request SME Consent]

Creates consent session in PENDING state.

SCHEMATIC 02 · SME PORTAL

Consent inspection

Borrower authorization screen

Requester: Demo Lending Bank

Scope: 12M Bank + GST Returns

Decision: [Approve] / [Reject]

Explicit gate before data pipeline fetch.

SCHEMATIC 03 · REVIEW VIEW

Risk dashboard

Underwriter synthesis panel

Overall score: 72 / 100 (Moderate)

Risk flags: 2 anomalies detected

Rec. limit: ₹5,76,000 preliminary

Shows breakdown rationale and anomalies.

END-TO-END USER JOURNEY RAIL

CREATE FILE (Bank) → REQUEST CONSENT (Bank) → APPROVE DATA ACCESS (SME) → FETCH AA DATA (Backend) → ANALYZE SIGNALS (Engine) → STRUCTURED REVIEW (Underwriter)

What we built today is useful, bounded, and ready for the next layer

IMPLEMENTED

- Next.js frontend; separate Bank and SME interfaces; FastAPI backend
- Application and consent APIs; financial-data APIs; deterministic risk engine
- Synthetic bank/GST datasets; multiple synthetic SME profiles: SME001 Baseline · SME002 Healthy · SME003 Higher-Risk; synthetic scenarios for demonstration
- Explainable results; preliminary credit-limit recommendation; frontend on Vercel and backend on Render

CURRENT LIMITATIONS

- Mock Account Aggregator; synthetic data; in-memory database; prototype scoring methodology
- No real banking or GST integrations; no real credit approval; no production authentication or authorization architecture

FUTURE EXTENSIONS

- Real AA integration; persistent database; authentication and roles; audit logs
- Stronger security and privacy controls; more financial features
- Historical-data-based validation; production-grade underwriting workflows